

1.13. Ohm's Law

This law applies to electric to electric conduction through good conductors and may be stated as follows :

The ratio of potential difference (V) between any two points on a conductor to the current (I) flowing between them, is constant, provided the temperature of the conductor does not change.

In other words,

$$\frac{V}{I} = \text{constant} \quad \text{or} \quad \frac{V}{I} = R$$

where R is the resistance of the conductor between the two points considered.

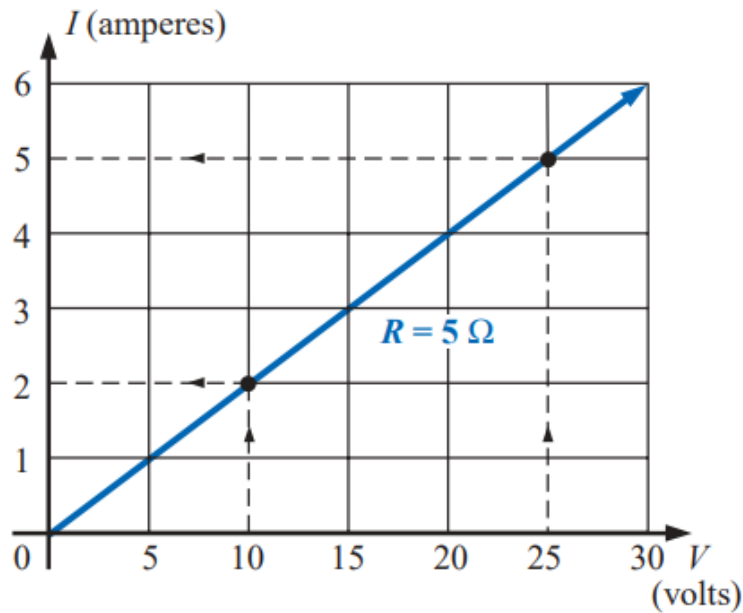


FIG. 4.6
Plotting Ohm's law.

1.14. Resistance in Series

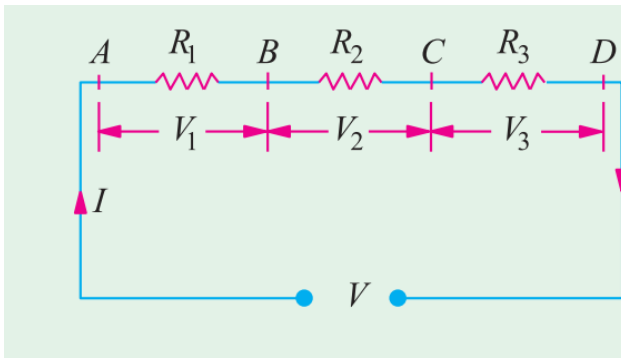


Fig. 1.12

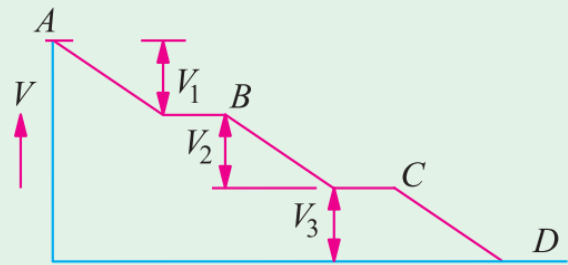


Fig. 1.13

$\therefore V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3$ —Ohm's Law
 But $V = IR$

where R is the equivalent resistance of the series combination.

$\therefore IR = IR_1 + IR_2 + IR_3$ or $R = R_1 + R_2 + R_3$

As seen from above, the main characteristics of a series circuit are :

1. same current flows through all parts of the circuit.
2. different resistors have their individual voltage drops.
3. voltage drops are additive.
4. applied voltage equals the sum of different voltage drops.
5. resistances are additive.
6. powers are additive.

1.15. Voltage Divider Rule

Since in a series circuit, same current flows through each of the given resistors, voltage drop varies directly with its resistance. In Fig. 1.14 is shown a 24-V battery connected across a series combination of three resistors.

Total resistance $R = R_1 + R_2 + R_3 = 12 \Omega$

According to Voltage Divider Rule, various voltage drops are :

$$V_1 = V \cdot \frac{R_1}{R} = 24 \times \frac{2}{12} = 4 \text{ V}$$

$$V_2 = V \cdot \frac{R_2}{R} = 24 \times \frac{4}{12} = 8 \text{ V}$$

$$V_3 = V \cdot \frac{R_3}{R} = 24 \times \frac{6}{12} = 12 \text{ V}$$

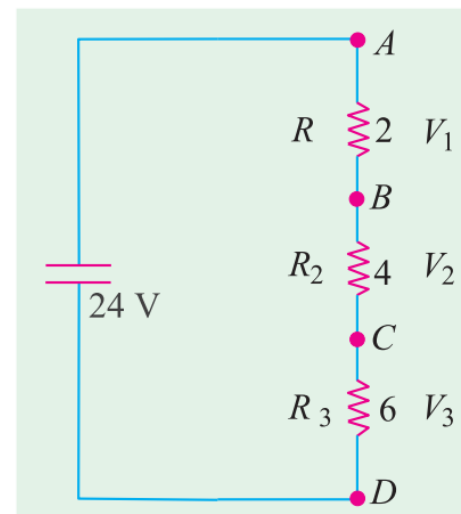
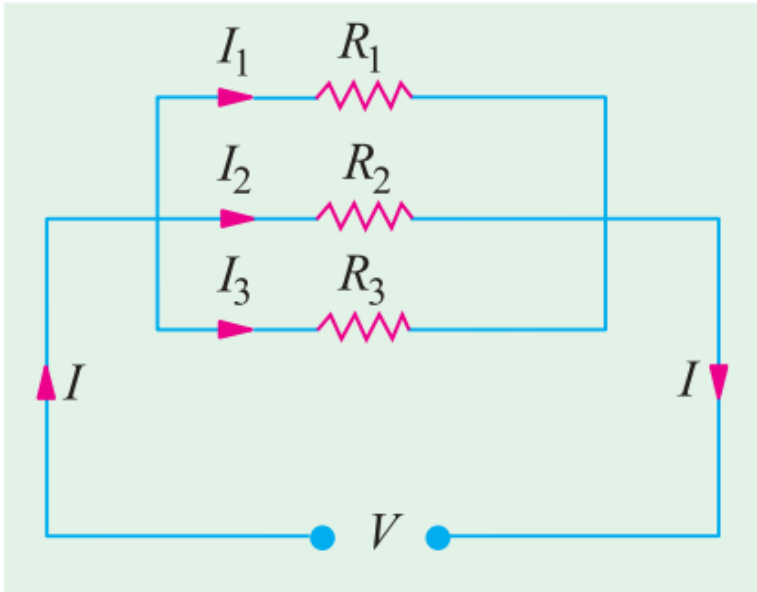


Fig.1.14

1.16. Resistances in Parallel



$$I = I_1 + I_2 + I_3 = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

Now,

$$I = \frac{V}{R} \text{ where } V \text{ is the applied voltage.}$$

R = equivalent resistance of the parallel combination.

$\therefore$

$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \quad \text{or} \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

The main characteristics of a parallel circuit are :

1. same voltage acts across all parts of the circuit
2. different resistors have their individual current.
3. branch currents are additive.
4. conductances are additive.
5. powers are additive.

Example 1.25. What is the value of the unknown resistor R in Fig. 1.16 if the voltage drop across the $500\ \Omega$ resistor is 2.5 volts? All resistances are in ohm. (Elect. Technology, Indore Univ.)

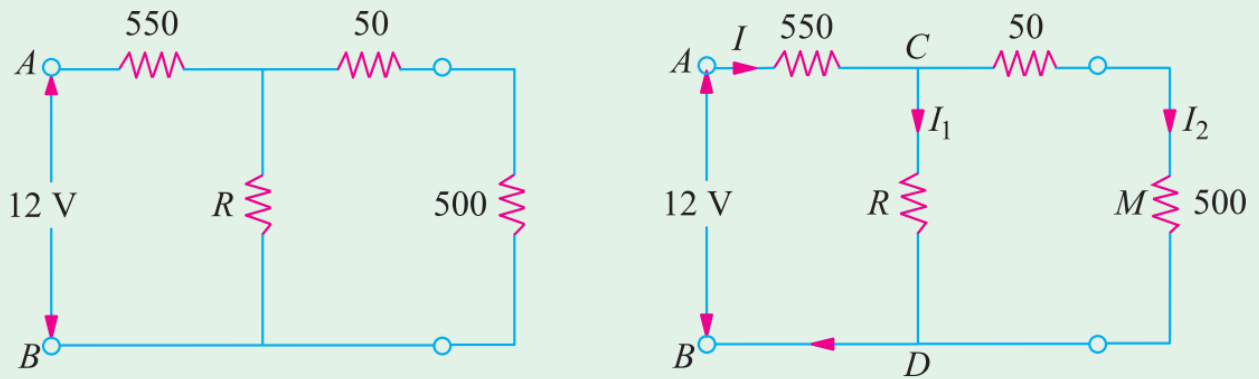


Fig. 1.16

Solution. By direct proportion, drop on $50\ \Omega$ resistance $= 2.5 \times 50/500 = 0.25\ \text{V}$

Drop across CMD or CD $= 2.5 + 0.25 = 2.75\ \text{V}$

Drop across $550\ \Omega$ resistance $= 12 - 2.75 = 9.25\ \text{V}$

$$I = 9.25/550 = 0.0168\ \text{A}, I_2 = 2.5/500 = 0.005\ \text{A}$$

$$I_1 = 0.0168 - 0.005 = 0.0118\ \text{A}$$

$$\therefore 0.0118 = 2.75/R; \quad R = 233\ \Omega$$

1.20. Short and Open Circuits

When two points of circuit are connected together by a thick metallic wire (Fig. 1.38), they are said to be *short-circuited*. Since ‘short’ has practically zero resistance, it gives rise to two important facts :

- (i) no voltage can exist across it because $V = IR = I \times 0 = 0$
- (ii) current through it (called short-circuit current) is very large (theoretically, infinity)

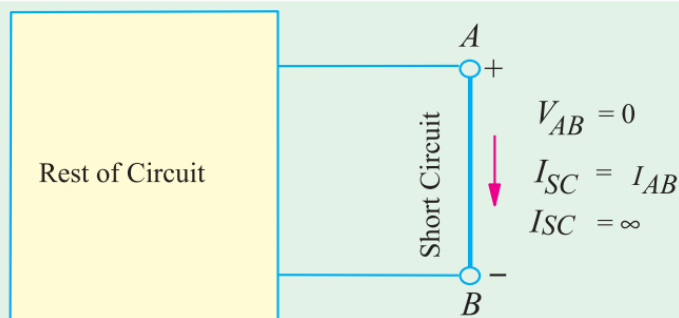


Fig. 1.38

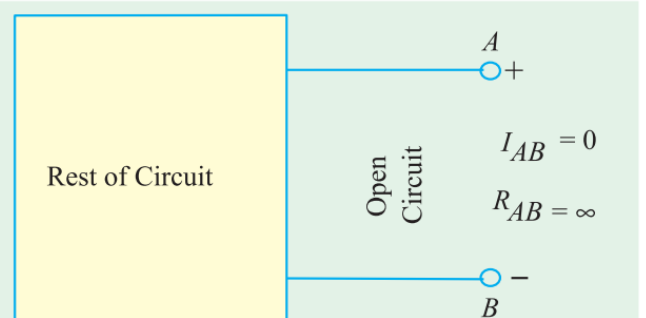


Fig. 1.39

Two points are said to be open-circuited when there is no direct connection between them (Fig. 1.39). Obviously, an ‘open’ represents a break in the continuity of the circuit. Due to this break

- (i) resistance between the two points is infinite.
- (ii) there is no flow of current between the two points.

Statement of Current Divider rule:

Statement: The electrical current entering the node of a parallel circuit is divided into the branches. Current divider formula is engaged to calculate the magnitude of divided current in the circuits.

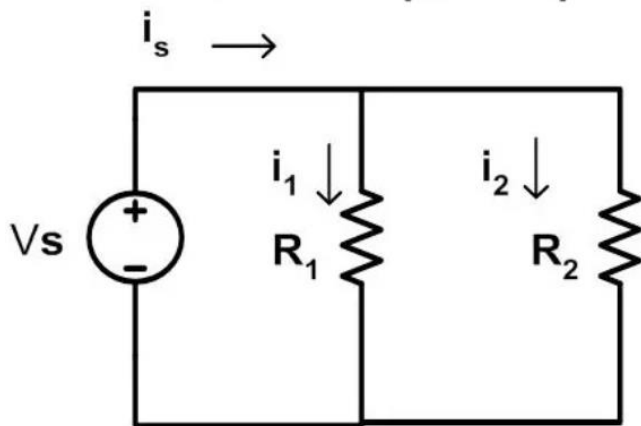
Current Divider Formula:

A common formula for the current I_X in a resistor R_X that is in parallel with a combination of other resistors of total resistance R_T is :

$$I_X = \frac{R_T}{R_X + R_T} I_T$$

Where I_T is the total current entered into the network of R_X in parallel with R_T . when R_T is composed of a parallel combination of resistors, say $R_1, R_2, \dots$ then the mutual of each resistor must be added to find the total resistance R_T :

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$



$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

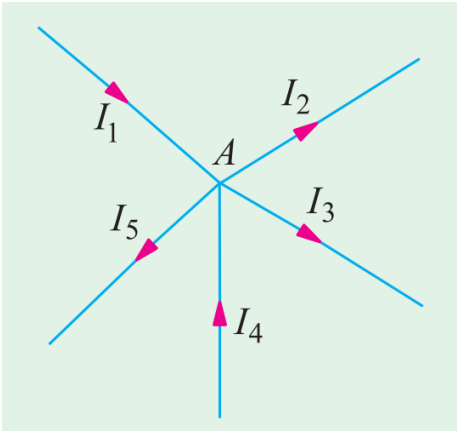
$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

1. Kirchhoff's Point Law or Current Law (KCL)

It states as follows :

in any electrical network, the algebraic sum of the currents meeting at a point (or junction) is zero.

Put in another way, it simply means that the total current *leaving* a junction is equal to the total current *entering* that junction. It is obviously true because there is no accumulation of charge at the junction of the network.



$$I_1 + (-I_2) + (-I_3) + (+I_4) + (-I_5) = 0$$

or

$$I_1 + I_4 - I_2 - I_3 - I_5 = 0 \quad \text{or} \quad I_1 + I_4 = I_2 + I_3 + I_5$$

or

incoming currents = outgoing currents

5.4 KIRCHHOFF'S VOLTAGE LAW

Note Fig. 5.11.

Kirchhoff's voltage law (KVL) states that the algebraic sum of the potential rises and drops around a closed loop (or path) is zero.

A **closed loop** is any continuous path that leaves a point in one direction and returns to that same point from another direction without leaving the circuit. In Fig. 5.12, by following the current, we can trace a continuous path that leaves point a through R_1 and returns through E without leaving the circuit. Therefore, $abcda$ is a closed loop. For us to be able to apply Kirchhoff's voltage law, the summation of potential rises and drops must be made in one direction around the closed loop.

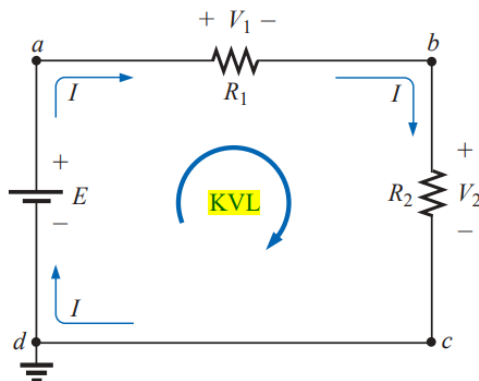


FIG. 5.12

Applying Kirchhoff's voltage law to a series dc circuit.

A plus sign is assigned to a potential rise ($-$ to $+$), and a minus sign to a potential drop ($+$ to $-$). If we follow the current in Fig. 5.12 from point a , we first encounter a potential drop V_1 ($+$ to $-$) across R_1 and then another potential drop V_2 across R_2 . Continuing through the voltage source, we have a potential rise E ($-$ to $+$) before returning to point a . In symbolic form, where Σ represents summation, $\oint$ the closed loop, and V the potential drops and rises, we have

$$\boxed{\Sigma_{\oint} V = 0} \quad \text{(Kirchhoff's voltage law in symbolic form)} \quad (5.8)$$

which for the circuit of Fig. 5.12 yields (clockwise direction, following the current I and starting at point d):

$$+E - V_1 - V_2 = 0$$

or

$$E = V_1 + V_2$$